MovieMind AI — Movie Recommendation System

Introduction

Movie recommendation systems help users discover relevant content from large collections. This project implements a hybrid recommendation system using movie genres and user ratings, with an interactive Streamlit interface.

Abstract

The system combines content-based filtering and collaborative filtering. TF-IDF represents movie genres as numerical features and cosine similarity measures similarity between movies. Collaborative filtering uses a user-item rating matrix and user similarity to estimate preference. A weighted hybrid score combines both signals and returns the top five unseen movies.

Tools Used

Python, Pandas, NumPy, Scikit-learn, TF-IDF, cosine similarity, Streamlit and Jupyter Notebook.

Steps Involved in Building the Project

- Prepare movie and rating CSV data.
- Clean and represent genres using TF-IDF.
- Calculate cosine similarity for content-based recommendations.
- Create a user-item rating matrix for collaborative filtering.
- Calculate user similarity and collaborative scores.
- Normalize and combine content and collaborative scores.
- Build a Streamlit UI for movie and genre preferences.
- Test the system with the included demo dataset.

Conclusion

MovieMind AI demonstrates the core workflow of an ML recommendation system and provides an easy-to-use web interface. The included dataset allows immediate execution, while the project structure supports replacing it with a larger MovieLens dataset for further experimentation.